PE3101P: Decision Theory and Social Choice

Tutorial 5 — Solutions

Instructor: Tomi Francis

Throughout, an **option set** is a finite, non-empty set of available options, and a **choice function** C gives, for every option set S, a non-empty subset $C(S)$ of S: what may be chosen from S. A person's or voter's ranking of options is an ordering: complete and transitive. In the gap-filling parts, the filled-in gaps are underlined.

Question 1

From any choice function C we can read off a ranking: $x \succsim y$ when $x \in C(\{x, y\})$. So $x \succ y$ when $C(\{x, y\}) = \{x\}$, and $x \sim y$ when $C(\{x, y\}) = \{x, y\}$. This ranking is complete, but it need not be transitive. An option is **maximal** in an option set, by a ranking, when nothing in the option set is strictly better than it. Sen's results relate consistency conditions on choice functions (Properties α, β and γ) to rankings. In this question we prove three facts about how they do so. Each part recalls the conditions it uses.

(a) In this part we will prove that if C satisfies expansion consistency (Property γ), then from every option set, C chooses every option that is maximal by the ranking read off from C. Recall:

> **Expansion consistency (Property γ)**: If x is chosen from an option set S_1, and x is chosen from an option set S_2, then x is chosen from the option set that combines them.
>
> $$[x \in C(S_1) \text{ and } x \in C(S_2)] \;\Rightarrow\; x \in C(S_1 \cup S_2)$$

Complete the proof. Fill each gap with one option set or the name of a condition.

Solution: Take any option set S, and any option x that is maximal in S. If $S = \{x\}$, then $C(S) = \{x\}$, since $C(S)$ is not empty. So suppose that S has other options too.

- Take any other option y in S. If x were not in $C(\{x, y\})$, then $C(\{x, y\})$ would be $\underline{\{y\}}$, since something is always chosen from $\{x, y\}$; and then $y \succ x$, which is false, since x is maximal in S. So $x \in C(\{x, y\})$.

- Combining the option sets $\{x, y\}$, for every other option y in S, gives the option set $\underline{S}$. So $x \in C(S)$, by $\underline{\text{expansion consistency (Property } \gamma)}$ (applied again and again).

(b) In this part, $\succsim$ is a complete and transitive ranking, and C chooses, from every option set, the maximal options by $\succsim$. We will show that C satisfies Property β:

> **Property β**: If x and y are both chosen from T, and $T \subset S$, then y is chosen from S if and only if x is.
>
> $$[x, y \in C(T) \text{ and } T \subset S] \;\Rightarrow\; [y \in C(S) \Leftrightarrow x \in C(S)]$$

Take option sets T and S with $T \subset S$, and options x and y that are both in $C(T)$. Suppose that $y \in C(S)$.

(i) Using the completeness of $\succsim$, show that $x \succsim y$.

Solution: Since $x \in C(T)$, x is maximal in T; and y is in T, so $y \succ x$ is false. By completeness, it follows that $x \succsim y$.[1] (By a symmetrical argument, we also have $y \succsim x$.)

Recall:

Transitivity: If $s \succsim t$ and $t \succsim u$, then $s \succsim u$. (So also: if $s \succ t$ and $t \succsim u$, then $s \succ u$.)

(ii) Show that nothing in S is strictly better than x.

Solution: Suppose, for contradiction, that some z in S has $z \succ x$. By (i), $x \succsim y$, so by transitivity, $z \succ y$. This contradicts $y \in C(S)$, since this requires that y is maximal in S.

So $x \in C(S)$. The same argument with x and y swapped shows that if $x \in C(S)$, then $y \in C(S)$. So C satisfies Property β.

(c) In this part, C is any choice function that satisfies Property β (as in (b)) and contraction consistency (Property α), and $\succsim$ is the ranking read off from C. We will prove that indifference ($\sim$) by this ranking is transitive. Recall:

Contraction consistency (Property α): If x is chosen from an option set S, and x is in a smaller option set T inside S, then x is chosen from T.

$$[x \in C(S) \text{ and } x \in T \subset S] \;\Rightarrow\; x \in C(T)$$

Complete the proof. Fill each gap with one option set or the name of a condition.

Solution: Take any three different options x, y and z with $x \sim y$ and $y \sim z$. So $C(\{x, y\}) = \{x, y\}$ and $C(\{y, z\}) = \{y, z\}$. Let $S = \{x, y, z\}$.

- x and y are both chosen from $\{x, y\}$, which is inside S. So if either of them is in $C(S)$, both are, by Property β. Likewise for y and z.
- So, since $C(S)$ is not empty, $C(S) = \{x, y, z\}$ (that is, S).
- So x and z are both chosen from $\{x, z\}$, by contraction consistency (Property α). So $x \sim z$.

For the second gap: since $C(S)$ is nonempty, at least one of x, y and z is chosen from S, and by the first step, whichever it is, so are the other two.

[1] In full: by completeness, $x \succsim y$ or $y \succsim x$. If $x \succsim y$ were false, we would have $y \succsim x$ without $x \succsim y$, which is $y \succ x$; and that is false.

Question 2

When two people are decisive over the same pair and disagree about it, each excludes one of its two states, so rights conflict even without Pareto. This question asks what else such a conflict needs. Recall:

> **Decisiveness**: A person is decisive over a pair of social states when, whichever of the two they strictly prefer, the other is not chosen from any option set containing both.

Say that a state in an option set is **excluded** when someone's decisiveness or right means that it is not chosen from that option set.

(a) Suppose that person 1 is decisive over one pair of social states, and person 2 over a different pair (the two pairs may share a state). Leave Pareto aside. Is there an option set, and a ranking for each person, under which every state in the option set is excluded by someone's decisiveness? If so, give one. If not, say why not.

Solution: No. A person's decisiveness excludes at most one state from an option set, and only if the whole of their pair is in it. So if one state is excluded, the option set has at least the two states of that person's pair, and one is left. And if two are excluded, both pairs are in the option set; since the pairs are different, it then has at least three states, and one is left. (This is why, in Sen's proof, only Case 1, where both people are decisive over the same pair, manages without Pareto.)

Gibbard gives each person a right over their own features instead, and rights of this kind can conflict without Pareto. Two people, 1 and 2, each choose whether to keep their own hair long (L) or short (S). A state is written with person 1's hair first: for example, SL is the state in which person 1's hair is short and person 2's is long. Each person ranks the four states LL, LS, SL and SS from best to worst, with no ties, and has a right over their own hair:

> **Rights over own hair**: If two states differ only in one person's hair, the one of the two that this person prefers excludes the other: the other is not chosen from any option set containing both.

So each person's right covers two pairs, not one: person 1's covers $\{\text{LL}, \text{SL}\}$ and $\{\text{LS}, \text{SS}\}$, and person 2's covers $\{\text{LL}, \text{LS}\}$ and $\{\text{SL}, \text{SS}\}$. Say that a person's preference about their own hair is **unconditional** when they prefer the same length for themselves whatever the other person's hair is. For example, person 1's preference is unconditional, with long hair preferred, when person 1 prefers LL to SL and also prefers LS to SS.

(b) In this part we show that one unconditional preference is enough to prevent a conflict. Suppose that person 1's preference about their own hair is unconditional; person 2's need not be. Show that from every option set whose states are among these four, at least one state is excluded by no right. (Start with the states of the option set in which person 1's hair is the length person 1 prefers.)

Solution: Say that person 1 prefers long hair for themselves (the short case is the same, with L and S swapped). Then person 1's right only ever excludes SL or SS, the states in which their hair is short. If the option set contains LL or LS, the only right that can exclude either of them is person 2's right over $\{\text{LL}, \text{LS}\}$, which excludes at most one of them. If it contains neither, it is inside $\{\text{SL}, \text{SS}\}$; person 1's right does nothing there, and person 2's right excludes at most one of the two. Either way, some state is excluded by no right.

Question 3

A course (PE3101P) has one place left. Rosie and Jim may each apply for it. Rosie decides first; Jim sees what she decided, and then decides. The result is one of three social states:

Choices	State	What happens
Rosie applies (whatever Jim does)	y	Rosie takes the place
Rosie does not apply, and Jim applies	z	Jim takes the place
Neither applies	x	Nobody takes the place

Whether Rosie applies is a personal matter for her, and whether Jim applies is a personal matter for him. Say that a person has a **right** to a feature of the result when some choice of theirs guarantees that feature, whatever the other person chooses.

(a) Fill in the table. In each cell, write "no" if the person has no right to the feature; if they have one, write the choice that guarantees it.

Solution:

	They take the place themselves	They do not take the place
Rosie	Apply	Do not apply
Jim	No	Do not apply

If Rosie applies, she takes the place whatever Jim does; if she does not apply, she does not take it. If Jim does not apply, he does not take the place whatever Rosie does. But no choice of Jim's guarantees that he takes the place: if Rosie applies, she takes it, even if he applies too.

Rosie does not really want the course, but she would rather take it herself than see Jim take it, since she doesn't want him to find out about contraction consistency. Jim wants the place most; failing that, he would rather it stayed empty than went to Rosie. So their rankings are:

	Rosie	Jim
Best	Nobody (x)	Jim (z)
Middle	Rosie (y)	Nobody (x)
Worst	Jim (z)	Rosie (y)

(b) Each of them knows the other's ranking, and each chooses what is best for themselves. Which state results?

Solution: y. If Rosie doesn't apply, Jim's best choice is to apply, since he prefers z to x. So if Rosie applies, she takes the place (y); if not, Jim does (z). Rosie prefers y to z, so her best choice is to apply, and the result is y.

(c) Suppose instead that Jim cannot apply, so that only Rosie decides. Which state results?

Solution: x. Rosie prefers x to y, so she doesn't apply.

Think of the state that results as what is chosen from the states that the procedure can lead to: from $\{x, y, z\}$ in (b), and from $\{x, y\}$ in (c). Recall:

> **Contraction consistency (Property α):** If x is chosen from an option set S, and x is in a smaller option set T inside S, then x is chosen from T.

(d) Does this way of choosing satisfy contraction consistency (Property α)? If not, give the states and option sets that show it.

Solution: No. y is chosen from the bigger option set $\{x, y, z\}$ in (b), and it is in the smaller option set $\{x, y\}$ in (c), but it is not chosen there.

On Sen's way of stating rights, Rosie is decisive over $\{x, y\}$, since these two states differ only in whether she applies. Recall:

Decisiveness: A person is decisive over a pair of social states when, whichever of the two they strictly prefer, the other is not chosen from any option set containing both.

(e) Does the result in (b) respect Rosie's decisiveness over $\{x, y\}$?

Solution: No: x was available, Rosie prefers it to y, and y is chosen.

(f) On the same way of stating rights, is Jim decisive over any pair of social states? If so, which? If not, say why not.

Solution: Yes: $\{x, z\}$, the two states that differ only in whether Jim applies. And unlike Rosie's, this decisiveness is respected, whatever the preferences. If Jim prefers x to z, he can make sure z doesn't happen, by not applying. If he prefers z to x, he can make sure x doesn't happen, by applying whenever Rosie hasn't. (Every other pair contains y, and y differs from x and from z in whether Rosie applies.)

(g) Now return to rights as choices. In (b), did either person's choice stop the other from getting a feature that, by your table in (a), the other had a right to?

Solution: No, and it couldn't have: this follows from what a right is. To have a right to a feature is to have a choice that guarantees it, whatever the other person chooses. So nothing the other person chooses can stop you getting a feature you have a right to. On this way of stating rights, people's rights can never conflict.

Recall:

Pareto: If everyone strictly prefers s to t, then t is not chosen from any option set containing s.

(h) Does the result in (b) satisfy Pareto?

Solution: No. Both prefer x to y, x was available, and y results. Leaving them free leads to a state that both rank below another.

Question 4

Recall that an ordering can be written as a list of **levels**, from top to bottom: options in the same level are tied, and an option in a higher level is strictly preferred to one in a lower level. A **ranking shift** takes one option out of its level and puts it back somewhere else in the list, as a new level of its own or into another level. Everything else stays as it was: every pair of options not involving the moved option is ranked as before.

(a) A voter's ordering is $a \succ b \sim c \succ d$, the list $\{a\}$, $\{b, c\}$, $\{d\}$. For each ordering below, can it be obtained from this one by a single ranking shift of b? Write "yes" or "no".

Solution: The test in each case: is every pair not involving b (that is, $\{a, c\}$, $\{a, d\}$ and $\{c, d\}$) ranked as before, $a \succ c$, $a \succ d$ and $c \succ d$?

	Ordering	By a ranking shift of b?
(i)	$a \sim b \succ c \succ d$	Yes: b joins a's level.
(ii)	$a \succ c \succ b \sim d$	Yes: b joins d's level.
(iii)	$a \succ b \succ d \succ c$	No: now $d \succ c$.
(iv)	$a \sim b \sim c \succ d$	No: now $a \sim c$.

In the rest of this question, there are three voters and four options, a, b, c and d, and every voter's ordering is strict (no ties). A social ordering function gives a social ordering $\succsim$, complete and transitive, for each profile of voters' orderings. The one here satisfies Unrestricted Domain and

Pareto: If every voter prefers x to y, then $x \succ y$.

IIA: Take any two profiles, and any two options x and y. If every voter ranks x against y the same way in both profiles, then society ranks x against y the same way in both.

Here are two profiles. P' is made from P by ranking shifts of b by voters 1 and 2; voter 3's ordering is the same in both.

P:

Voter	Ordering
1	$b \succ a \succ c \succ d$
2	$c \succ d \succ a \succ b$
3	$c \succ b \succ d \succ a$

P':

Voter	Ordering
1	$a \succ b \succ c \succ d$
2	$c \succ b \succ d \succ a$
3	$c \succ b \succ d \succ a$

At P, society's ordering is $c \succ d \succ a \succ b$. We work out society's ordering at P', one pair at a time.

(b) (i) For which pairs of options does IIA carry society's ranking over from P to P'?

Solution: $\{a, c\}$, $\{a, d\}$, $\{b, c\}$ and $\{c, d\}$.

Voter 3 is the same in both profiles, and only b moved, so the pairs not involving b ($\{a, c\}$, $\{a, d\}$, $\{c, d\}$) are ranked the same way by every voter at P and P'. Of the pairs involving b, check voters 1 and 2:

Pair	Voter 1, P to P'	Voter 2, P to P'	Same for every voter?
$\{a, b\}$	$b \succ a$ to $a \succ b$	$a \succ b$ to $b \succ a$	No
$\{b, c\}$	$b \succ c$ to $b \succ c$	$c \succ b$ to $c \succ b$	Yes
$\{b, d\}$	$b \succ d$ to $b \succ d$	$d \succ b$ to $b \succ d$	No

$\{b, c\}$ counts, although b moved, because neither move took b past c. So IIA carries across from P to P': $c \succ a$, $d \succ a$, $c \succ b$ and $c \succ d$.

(ii) For each of the other pairs, which condition, or which property of a social ordering, fixes society's ranking at P'?

Solution:

– $\{b, d\}$: **Pareto**. At P', every voter prefers b to d, so $b \succ d$.

– $\{a, b\}$: **transitivity** of the social ordering. From $b \succ d$ (just shown) and $d \succ a$ (from (i)), $b \succ a$.

(iii) So what is society's ordering at P'?

Solution: $c \succ b \succ d \succ a$. This agrees with all six pairs: $c \succ b$, $c \succ d$, $c \succ a$ (from (i)), $b \succ d$, $b \succ a$ (from (ii)) and $d \succ a$ (from (i)).

Question 5

This question walks through the last stage of the proof of Arrow's Theorem, with three voters, 1, 2 and 3, and three options, a, b and c. Part (a) finds the pivotal voter for b. Part (b) shows, for any social ordering function satisfying the conditions below, that the pivotal voters for the other two options must be the same voter as the pivotal voter for b. Part (c) returns to the function in (a) and shows that its pivotal voter for b is a dictator.

Throughout, a social ordering function gives a social ordering $\succsim$, complete and transitive, for each profile, and satisfies Unrestricted Domain and

Pareto: If every voter prefers x to y, then $x \succ y$.

IIA: Take any two profiles, and any two options x and y. If every voter ranks x against y the same way in both profiles, then society ranks x against y the same way in both.

To find the **pivotal voter** for an option, start from a profile in which every voter puts that option strictly at the bottom. Let the voters, one at a time and in some fixed order, each move that option to the top of their own ranking, changing nothing else. The pivotal voter is the voter whose move first makes society put the option strictly at the top.

(a) Here the voters move b to the top of their rankings in the order voter 3, then voter 1, then voter 2:

Profile	Voter 1	Voter 2	Voter 3	
P_0	$a \succ c \succ b$	$c \succ a \succ b$	$a \succ c \succ b$	
P_1	$a \succ c \succ b$	$c \succ a \succ b$	$b \succ a \succ c$	(voter 3 has moved b to the top)
P_2	$b \succ a \succ c$	$c \succ a \succ b$	$b \succ a \succ c$	(voter 1 has moved b to the top)
P_3	$b \succ a \succ c$	$b \succ c \succ a$	$b \succ a \succ c$	(voter 2 has moved b to the top)

At P_1, society's ordering is $a \succ c \succ b$, and at P_2 it is $b \succ a \succ c$. Which voter is the pivotal voter for b?

Solution: Voter 1. At P_0, society puts b strictly at the bottom, by Pareto (every voter prefers a and c to b). At P_1, after voter 3's move, b is still at the bottom (given), and at P_2 it is strictly at the top for the first time. The move between P_1 and P_2 was voter 1's, so voter 1 is the pivotal voter.

Recall:

> **Decisiveness**: A voter i is decisive over a pair $\{x, y\}$ when, at every profile, whichever of the two they prefer, society prefers too (where $x \succ_i y$ means that voter i strictly prefers x to y): if $x \succ_i y$, then $x \succ y$; and if $y \succ_i x$, then $y \succ x$.
>
> **Pivotal voters are decisive**: For any social ordering function satisfying Unrestricted Domain, Pareto and IIA, with finitely many voters and at least three options, the pivotal voter for an option, found in this way, is decisive over every pair of options not involving that option.

Write $V(a)$, $V(b)$ and $V(c)$ for the pivotal voters for a, b and c. So $V(a)$ is decisive over $\{b, c\}$, $V(b)$ over $\{a, c\}$, and $V(c)$ over $\{a, b\}$.

(b) In this part, take any social ordering function satisfying Unrestricted Domain, Pareto and IIA whose pivotal voter for b is voter 2. Nothing is known about who $V(a)$ and $V(c)$ are.

(i) Suppose, for contradiction, that $V(a)$ is not voter 2. Give a single profile that shows this is impossible, whichever of voters 1 and 3 is $V(a)$, and whoever $V(c)$ is: at your profile, the decisiveness of $V(a)$, $V(b)$ and $V(c)$ must force society's ranking not to be transitive. Show how society must rank each pair, and say whose decisiveness forces it.

Solution: The idea, as in the lecture: give each pivotal voter the preference on their pair that makes society go round in a cycle.

Voter	Decisive over	Give them
$V(c)$ (anyone)	$\{a, b\}$	$a \succ b$
$V(a)$ (voter 1 or voter 3)	$\{b, c\}$	$b \succ c$
$V(b)$ (voter 2)	$\{a, c\}$	$c \succ a$

Since we don't know who $V(c)$ is, every voter gets $a \succ b$; since we don't know which of voters 1 and 3 is $V(a)$, both get $b \succ c$. So voter 2 has $c \succ a \succ b$, and voters 1 and 3 have $a \succ b \succ c$:

Voter	Ordering
1	$a \succ b \succ c$
2	$c \succ a \succ b$
3	$a \succ b \succ c$

Pair	Society	Forced by
$\{a, b\}$	$a \succ b$	$V(c)$, whoever it is
$\{a, c\}$	$c \succ a$	$V(b)$, voter 2
$\{b, c\}$	$b \succ c$	$V(a)$, voter 1 or voter 3

By Unrestricted Domain, this profile is allowed; but society would have $a \succ b$, $b \succ c$ and $c \succ a$, a cycle, which is not an ordering. So $V(a)$ is voter 2. (Since every voter has $a \succ b$, Pareto also forces $a \succ b$. The reverse cycle works too: voter 2 with $b \succ a \succ c$, and voters 1 and 3 with $c \succ b \succ a$.)

(ii) By (i), $V(a)$ is voter 2. Suppose, for contradiction, that $V(c)$ is not voter 2. Does your profile from (i) also show that this is impossible? If it does, show how society must rank each pair at it, and say whose decisiveness forces it. If not, say why not, and give a profile that does.

Solution: No. Now voter 2 is $V(a)$ as well as $V(b)$, so voter 2 settles both $\{b, c\}$ and $\{a, c\}$; and in (i) every voter, voter 2 included, had to agree on $\{a, b\}$. So at

that profile society just follows voter 2's ordering, which is transitive: with the profile above, $c \succ a \succ b$.

A profile that does show it: swap a and c throughout the profile from (i). Swapping them turns the argument about $V(a)$ into the same argument about $V(c)$.

Voter	Ordering
1	$c \succ b \succ a$
2	$a \succ c \succ b$
3	$c \succ b \succ a$

Pair	Society	Forced by
$\{a, b\}$	$b \succ a$	$V(c)$, voter 1 or voter 3
$\{a, c\}$	$a \succ c$	$V(b)$, voter 2
$\{b, c\}$	$c \succ b$	$V(a)$, whoever it is

Society would have $b \succ a$, $a \succ c$ and $c \succ b$: a cycle again. So $V(c)$ is voter 2.

(c) Return to the social ordering function in (a).

(i) Which voter is its dictator (a voter who is decisive over every pair)? Give your reason, using (a) and the argument of (b).

Solution: Voter 1. By (a), $V(b)$ is voter 1. The argument of (b) used nothing about voter 2 except that voter 2 is $V(b)$, and this function satisfies the same conditions; so, with voter 1 in voter 2's place, it shows that $V(a)$ and $V(c)$ are voter 1 too. The profiles are:

- for $V(a)$: voter 1 with $c \succ a \succ b$, and voters 2 and 3 with $a \succ b \succ c$;
- for $V(c)$: voter 1 with $a \succ c \succ b$, and voters 2 and 3 with $c \succ b \succ a$.

So voter 1 is decisive over $\{b, c\}$ (as $V(a)$), over $\{a, c\}$ (as $V(b)$) and over $\{a, b\}$ (as $V(c)$): over every pair.

(ii) At the profile R below, for which pairs of options does the dictator's decisiveness settle how society ranks them? For any pair it does not settle, say why not.

R:

Voter	Ordering
1	$a \sim b \succ c$
2	$b \succ c \succ a$
3	$c \succ a \succ b$

Solution:

- **Settled:** $\{a, c\}$ and $\{b, c\}$. Voter 1 strictly prefers a to c and b to c, so society has $a \succ c$ and $b \succ c$.
- **Not settled:** $\{a, b\}$. Voter 1 is indifferent between a and b, and decisiveness says only what society does when the voter strictly prefers one of the two.

(iii) Suppose the voters had instead moved b to the top in the order voter 1, then voter 2, then voter 3, starting from P_0. Which voter would be the pivotal voter for b?

Solution: Voter 1, the dictator. Voter 1 is decisive over all the pairs involving b, so society moves b to the top exactly when voter 1 does, whatever order the voters move in.